

Discrete behaviors often emerge from continuous physical systems.

Many physical systems governed by continuous equations—like optical cavities, electronic loops, and microwave resonators—exhibit surprisingly discrete modes of oscillation.

This raises a fundamental question: What physical mechanism selects these discrete states from a continuum of possibilities?

This presentation outlines a purely classical route to mode discreteness, driven by the internal dynamics of the system itself, without relying on microscopic quantization assumptions.



Delay-induced pattern formation provides a generic classical route to mode discreteness.

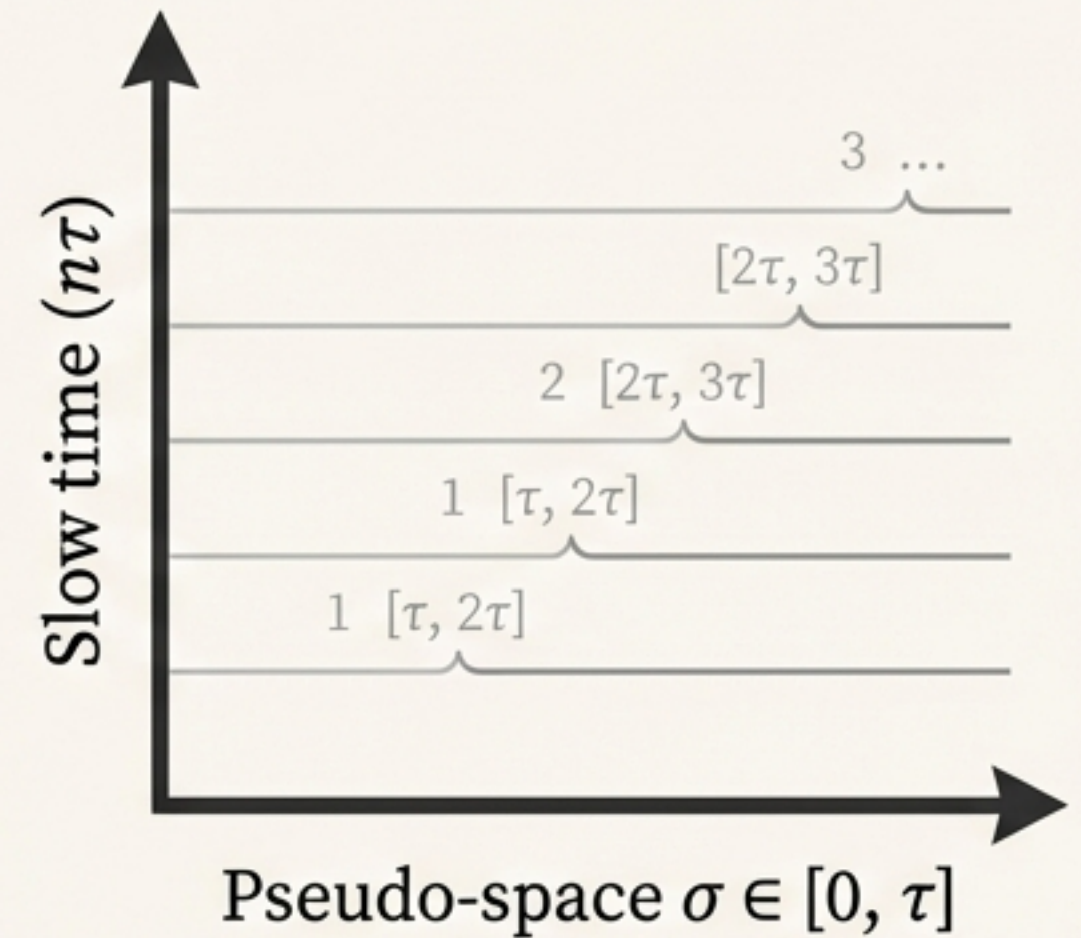
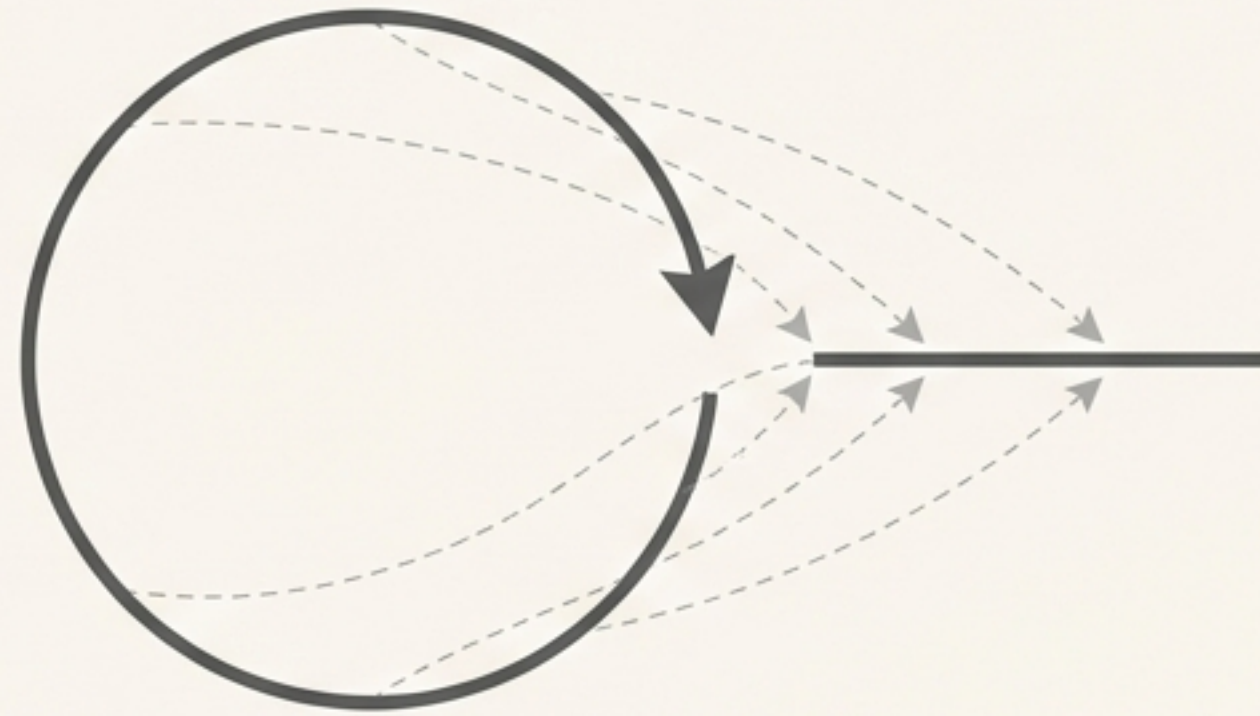
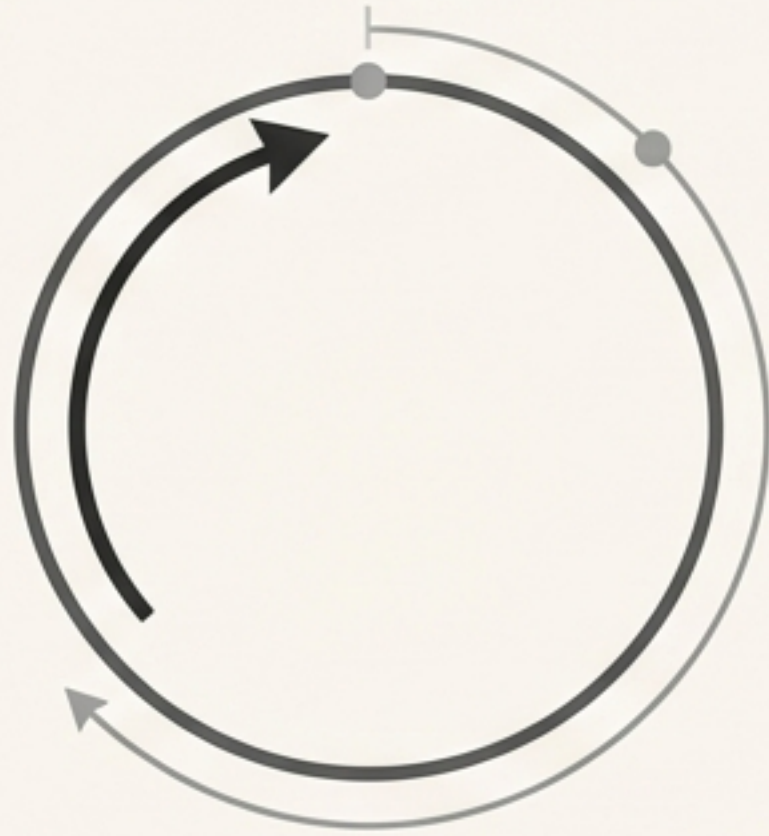


In any nonlinear loop with a finite propagation time, feedback can force the circulating field to self-organize into stable, discrete plateau-like patterns.

- This process yields quantized-like circulation modes selected purely by classical delay dynamics.
- The mechanism is physically transparent and offers a unifying framework for understanding mode selection in a wide range of nonlinear ring systems.

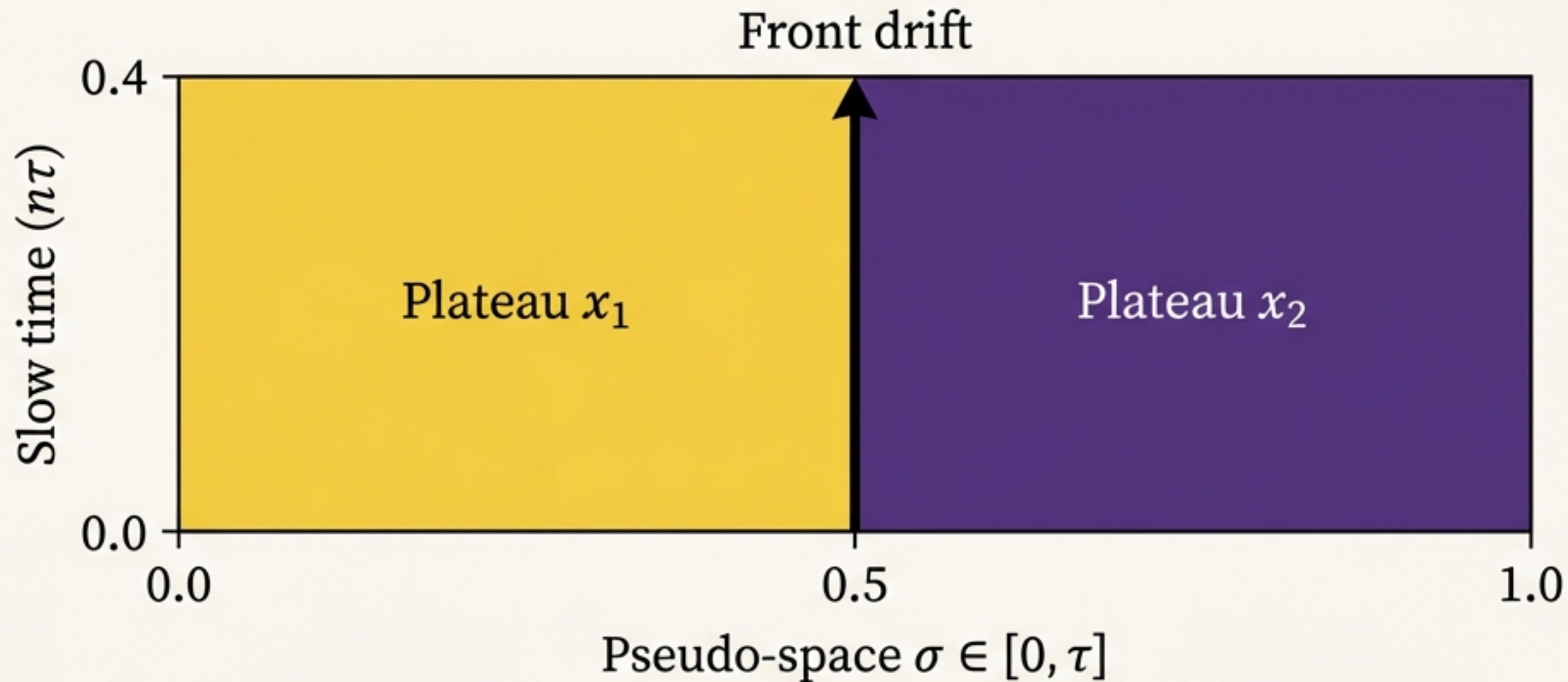
The long-delay regime allows for a powerful spatio-temporal interpretation.

In systems where the delay time τ is much larger than the system's relaxation time ε ($\tau/\varepsilon \gg 1$), the delay interval can be treated as a *pseudo-spatial* domain.



time delay $\rightarrow$ spatial extent

Conceptually, the system organizes into distinct plateau domains separated by fronts.



Takeaway: This schematic illustrates how the system can self-organize into stable, spatially distinct regions within the pseudo-space defined by the delay.

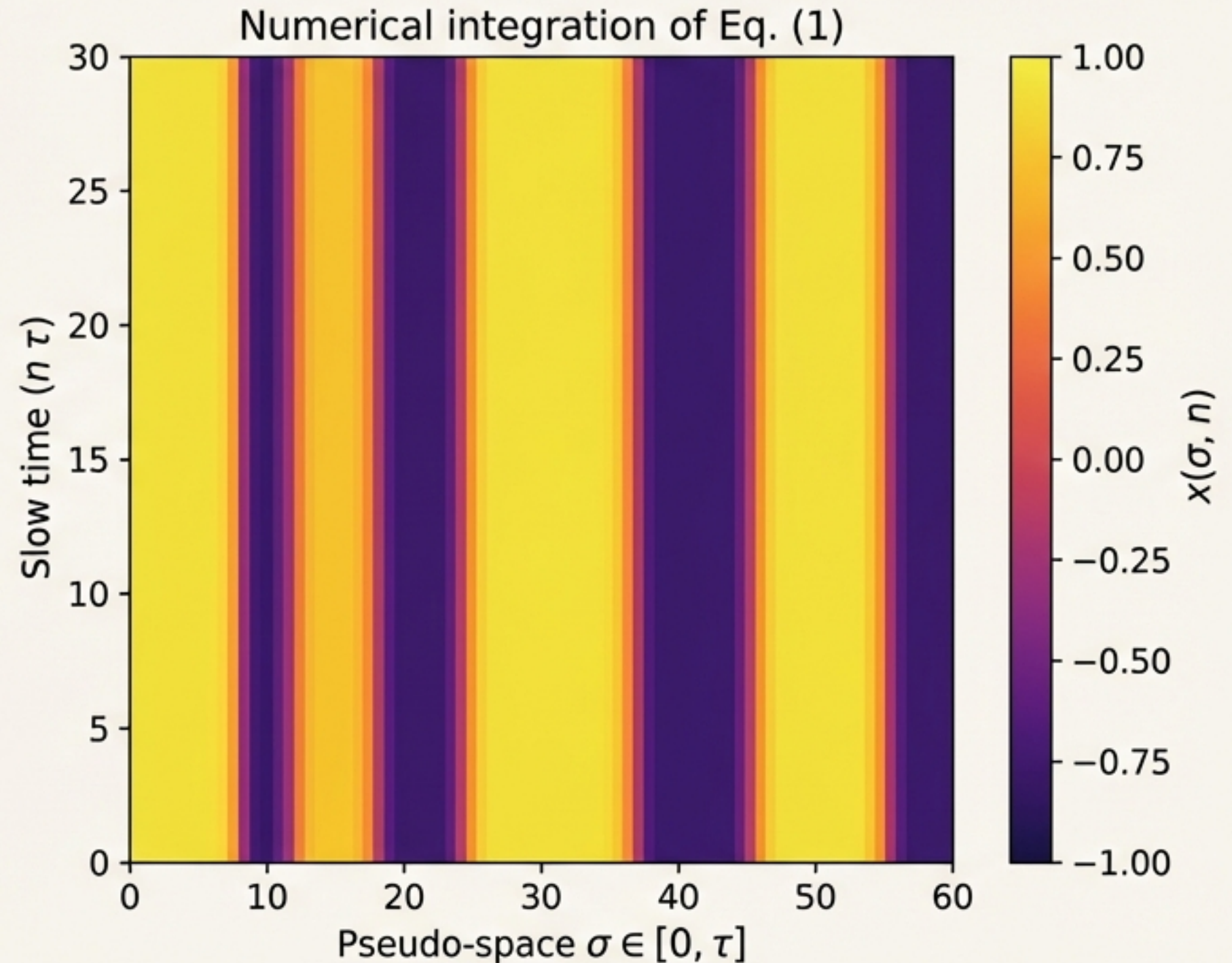
Numerical simulations confirm this spontaneous self-organization into discrete modes.

Direct numerical integration of a minimal delay model shows that a uniform state is unstable.

The system spontaneously breaks symmetry and settles into a stable pattern of alternating plateaus and sharp domain walls, confirming the conceptual picture.

$$\epsilon \dot{x}(t) = -x(t) + \mu x(t - \tau) - x^3(t - \tau)$$

Parameters used: $\epsilon = 0.01$, $\mu = 1.5$, $\tau = 60$.



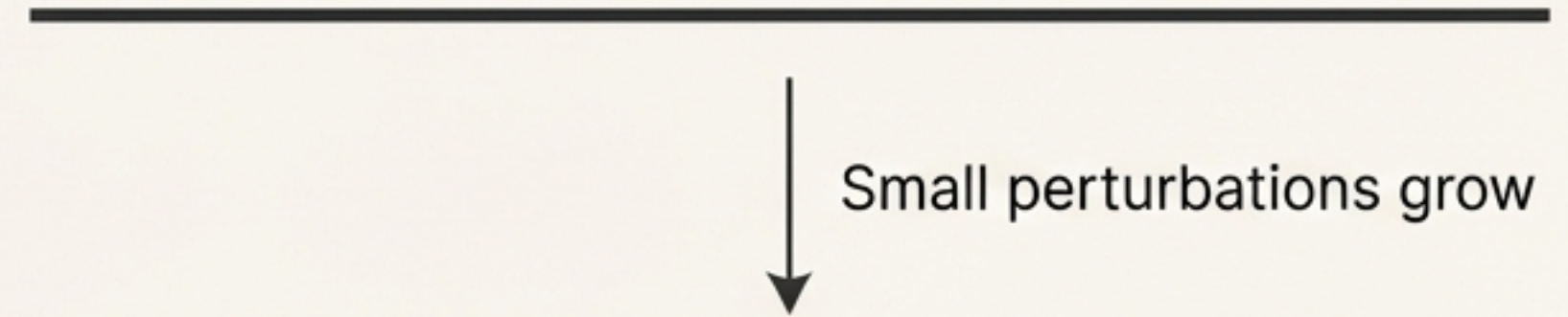
The patterns are selected by a delay-induced modulational instability.

Linear stability analysis of the trivial state $x = 0$ reveals the conditions for instability.

In the long-delay limit ($\tau \gg \varepsilon$), the eigenvalues λ_n of the system organize into quasi-continuous branches.

This leads to a modulational instability, where small perturbations grow and break the uniformity of the circulating field, initiating pattern formation.

Stage 1: Uniform State



Stage 2: Modulational Instability



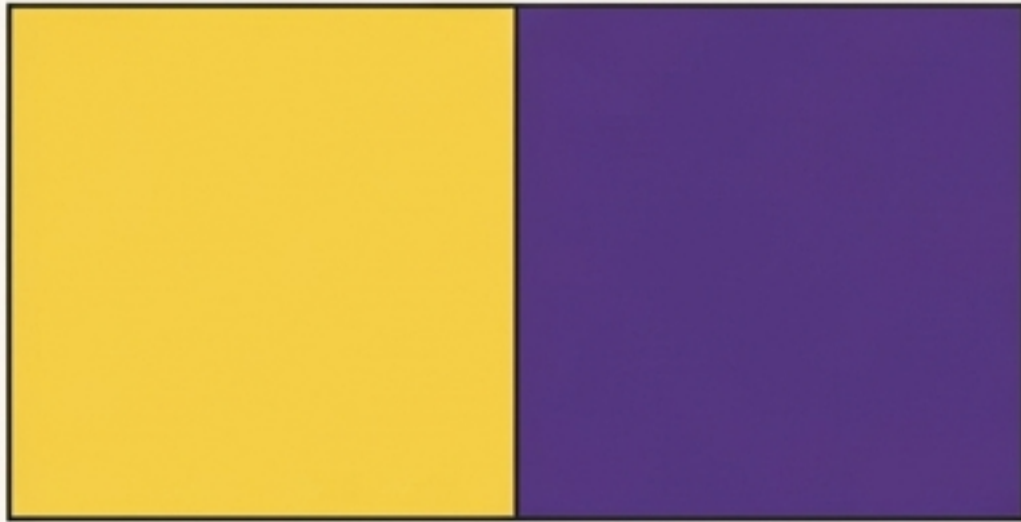
$$\epsilon\lambda + 1 = \mu e^{-\lambda\tau}$$

This equation determines the stability of the system. Its solutions with positive real parts correspond to unstable, growing modes.

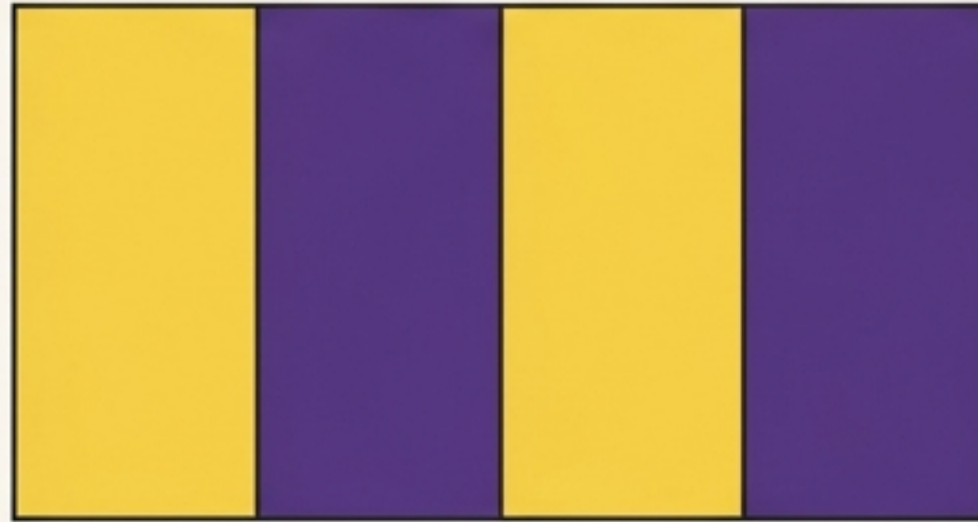
The system is multistable: distinct plateau configurations can coexist.

For the same set of system parameters (μ, τ, ε) , the dynamics can support multiple different stable patterns.

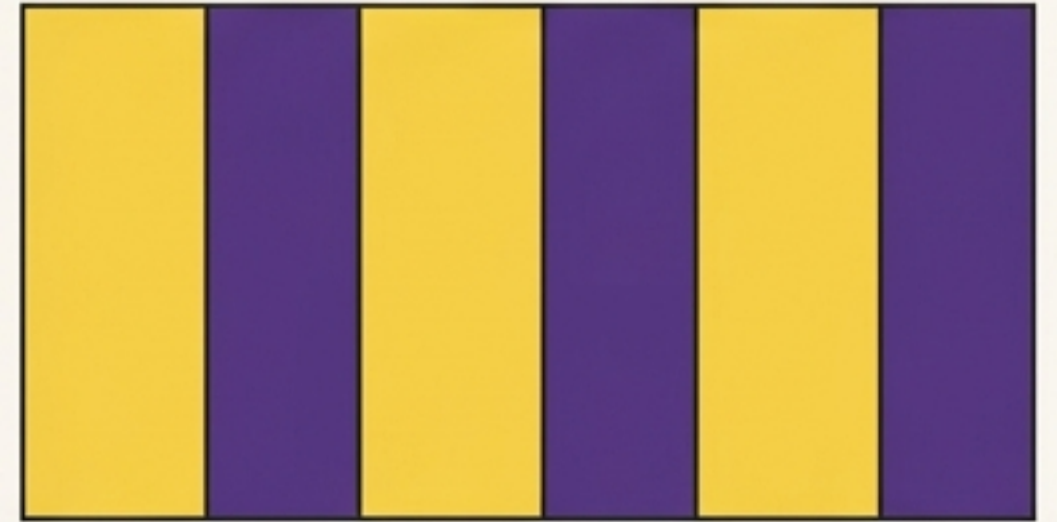
Mode 1



Mode 2



Mode 3



Implication: These distinct, coexisting configurations behave as different, discrete “circulation mode numbers,” selected by the initial conditions.

Understanding these modes requires a two-part explanation: selection and persistence.

1. SELECTION



What mechanism creates and selects the number of plateaus?

Delay-induced modulational instability. It breaks the symmetry and carves the continuum into preferred patterns.

2. PERSISTENCE



Once a pattern is formed, why is it so robust and long-lived?

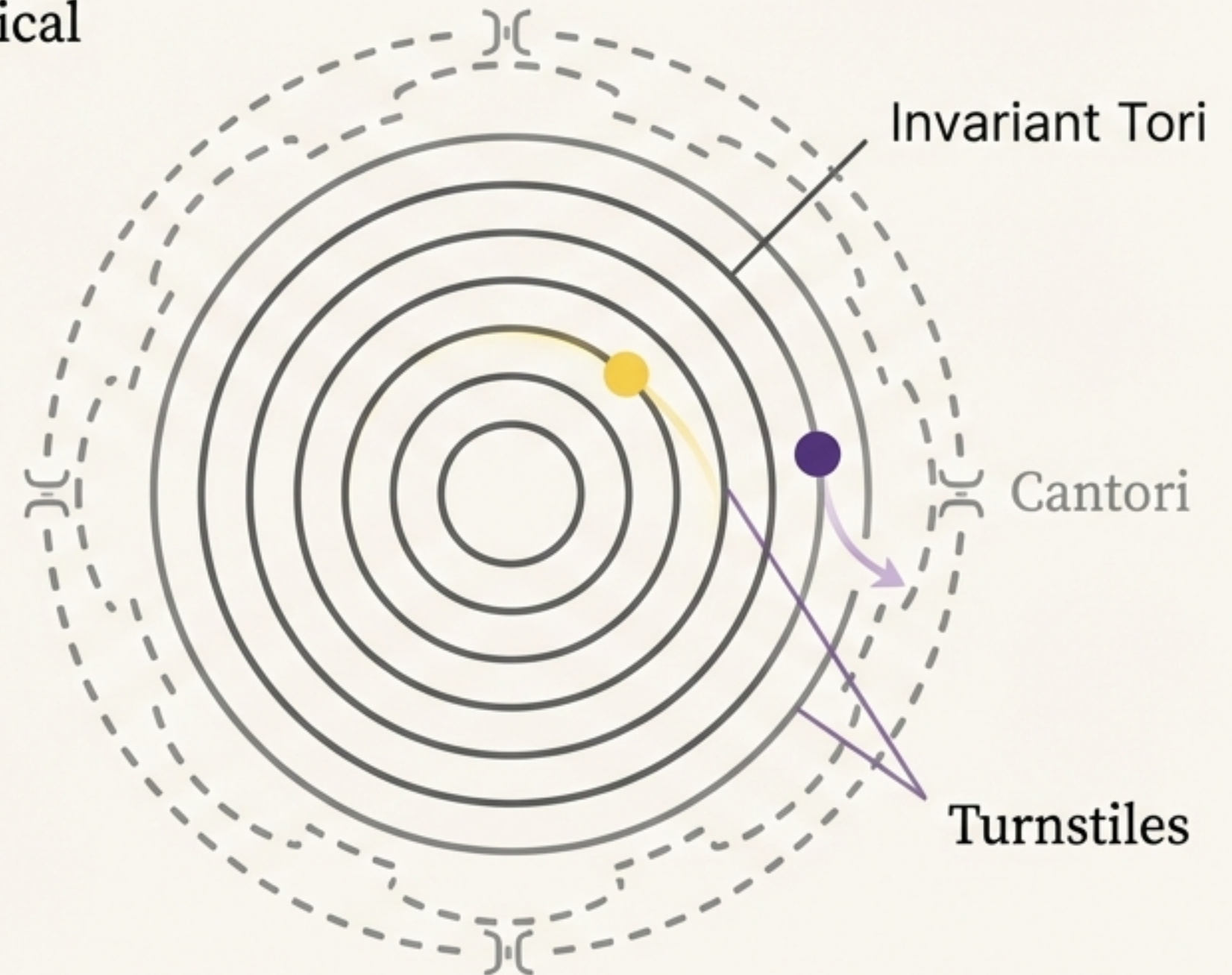
A separate mechanism related to the system's underlying topology provides stability.

The first question is answered by the dynamics we've shown. The second requires an analogy to transport barriers in Hamiltonian systems.

The persistence of modes can be understood through an analogy to topological transport barriers.

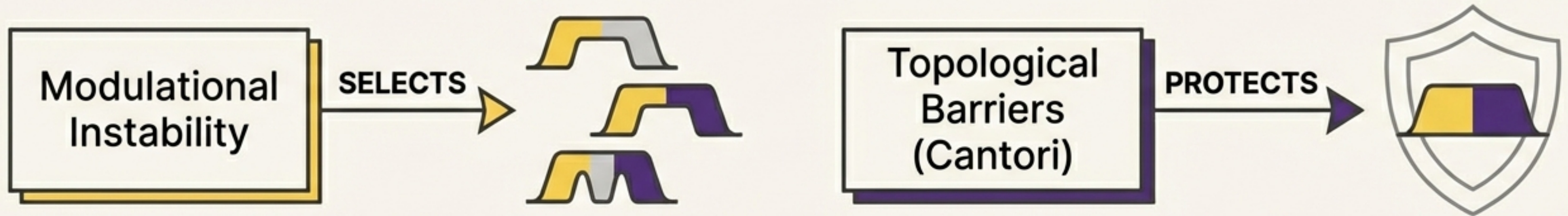
The circulating fields in ring systems share topological features with Hamiltonian flows on a torus.

- **Invariant Tori:** In ideal Hamiltonian systems, these act as absolute barriers, trapping trajectories.
- **Cantori:** Under perturbation, tori break into fractal, “leaky” remnants called cantori. They are no longer absolute barriers but still strongly suppress transport between regions.
- **Turnstiles:** These structures act as discrete channels or gates that mediate the slow, rare flux transfer through the gaps in the cantori.



**Selection determines which modes appear;
topology determines which modes persist.**

“Delay-induced pattern formation determines which modes **appear**, while toroidal Hamiltonian topology determines which modes **persist**.”

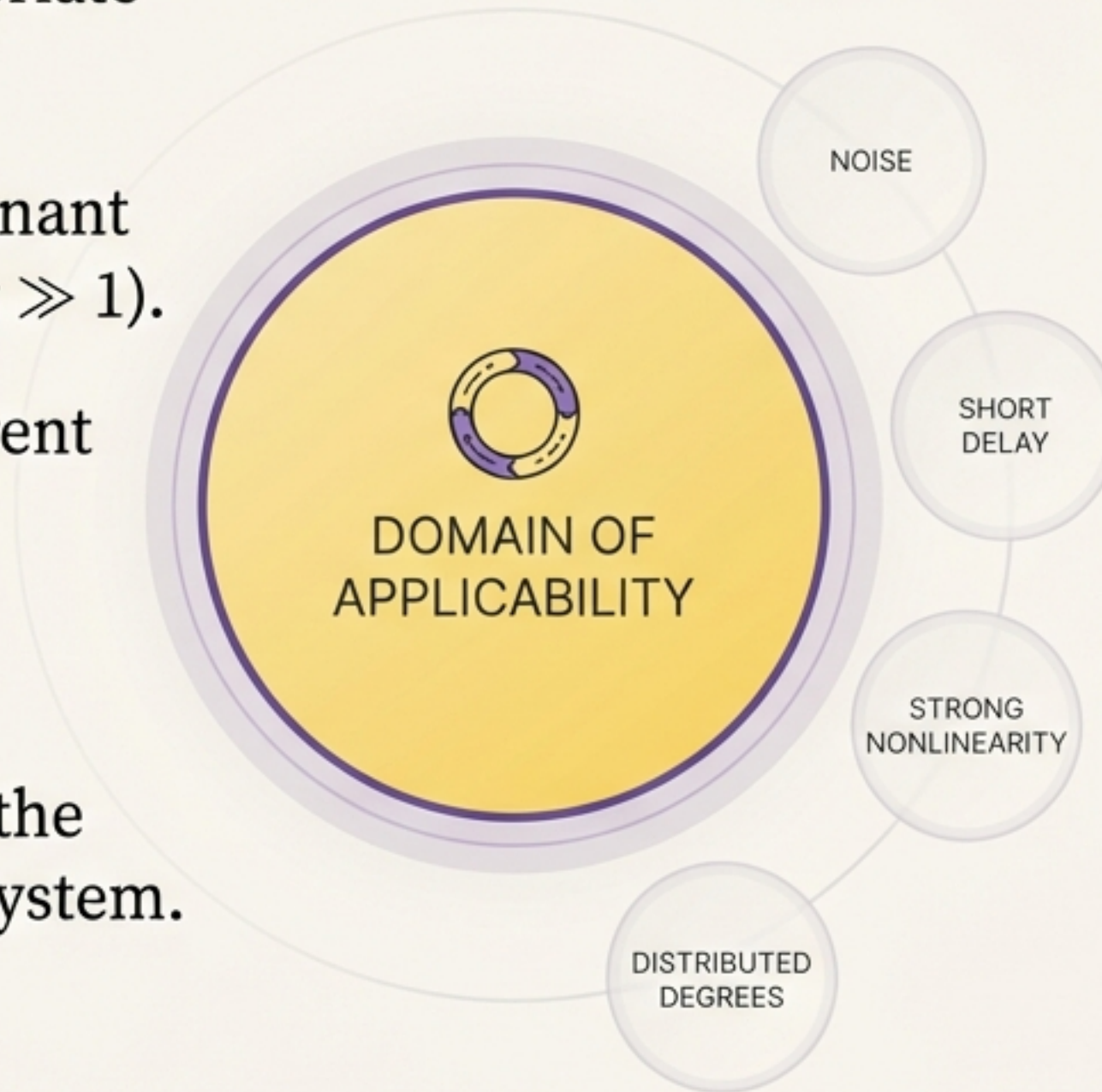


This two-part framework provides a complete classical explanation for the emergence and stability of discrete modes in nonlinear ring systems.

This spatio-temporal interpretation is most powerful in the long-delay regime.

The framework is most appropriate when:

1. The delay τ defines a dominant propagation timescale ($\tau/\epsilon \gg 1$).
2. The dynamics admit coherent fronts or domains on the pseudo-space $\sigma \in [0, \tau]$.
3. Boundary conditions are effectively periodic due to the closed-loop nature of the system.

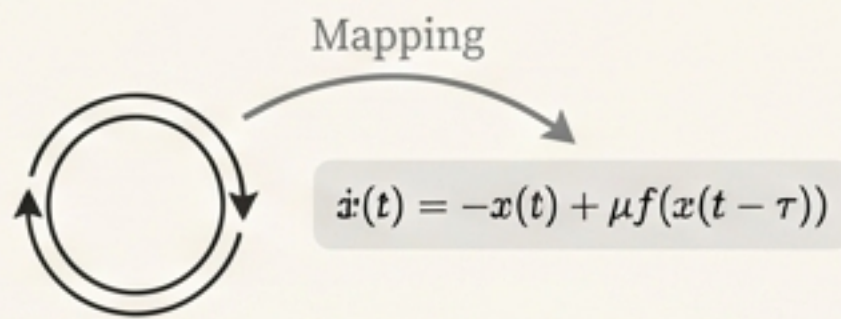
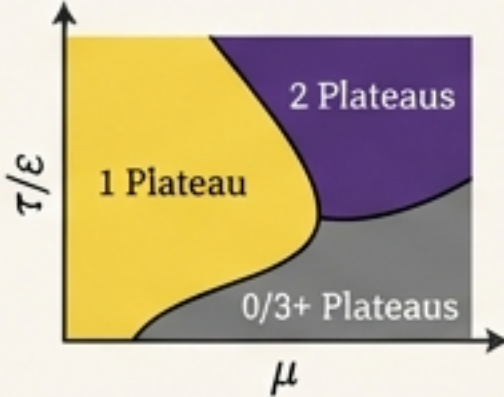
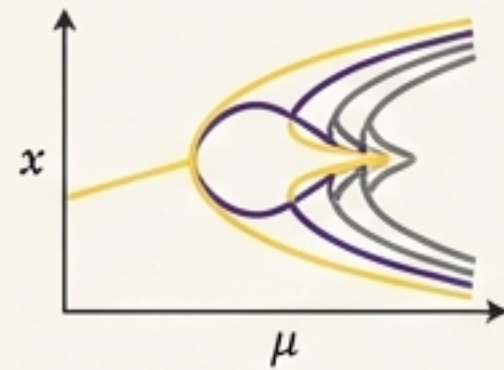


The correspondence may weaken in cases of:

- Strong stochasticity or noise.
- Highly non-smooth nonlinearities.
- When other distributed degrees of freedom dominate the dynamics.

Future work can quantitatively strengthen and test this framework.

Three immediate extensions are proposed:



1. **Bifurcation Analysis:** Numerically continue the solution branches to map out the full multistability regions and reveal their global structure (e.g., collapsed snaking).
2. **Data-Driven Mapping:** Systematically generate a phase diagram showing the number of stable plateaus as a function of the key parameters $(\mu, \tau/\varepsilon)$.
3. **Experimental Connection:** Develop a system-specific mapping from a physical ring (e.g., an optical or electronic loop) to the effective delay model parameters, enabling falsifiable predictions.

Appendix: The dynamics near onset reduce to the Complex Ginzburg-Landau Equation.

Context

Near the Hopf bifurcation threshold ($\mu \approx 1$), a multiple-scale expansion can be used to derive a simplified envelope equation.

Decomposition

The solution can be written as $x(t) = A(\zeta, \theta)e^{i\omega t} + c.c.$, where A is a slowly varying complex envelope.

The CGLE

The dynamics of the envelope A are governed by the Complex Ginzburg-Landau Equation (CGLE):

$$\partial_{\theta} A = \mu A + (1 + i\alpha)\partial_{\zeta}^2 A - (1 + i\beta)|A|^2 A$$

Significance

This reduction demonstrates that the delay-induced pattern formation belongs to a well-understood universality class of spatio-temporal dynamics, connecting it to a vast body of literature on pattern formation.